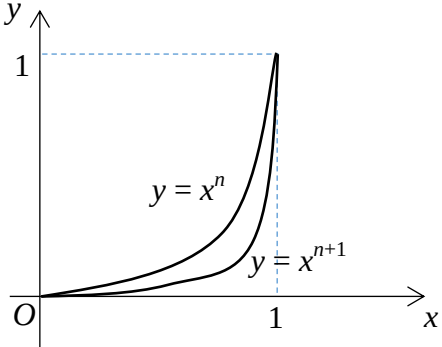
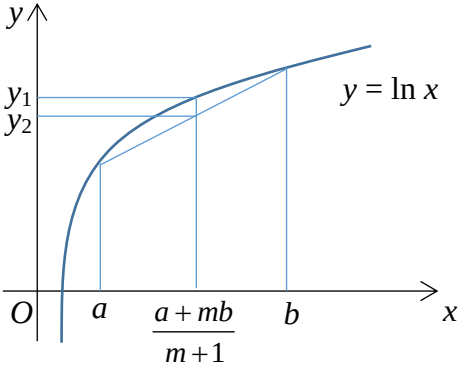


2015 RI H3 Mathematics Preliminary Examinations

1(a)	
(b) (i)	 From the graph, $y = \ln x$ is concave. Thus $y_2 \leq y_1 \Rightarrow \frac{\ln a + m \ln b}{m+1} \leq \ln \left(\frac{a + mb}{m+1} \right) \text{ with equality when } a = b.$
(ii)	$\frac{\ln a + m \ln b}{m+1} \leq \ln \left(\frac{a + mb}{m+1} \right)$ $\Rightarrow \ln \left(ab^m \right)^{\frac{1}{m+1}} \leq \ln \left(\frac{a + mb}{m+1} \right)$ Since $\ln$ is an increasing function, $\left(ab^m \right)^{\frac{1}{m+1}} \leq \left(\frac{a + mb}{m+1} \right)$ $\Rightarrow ab^m \leq \left(\frac{a + mb}{m+1} \right)^{m+1}.$
(iii)	We have $x^n(1-x) = n^n \left(\frac{x}{n} \right)^n (1-x) \leq n^n \left(\frac{(1-x) + n \left(\frac{x}{n} \right)}{n+1} \right)^{n+1} = \frac{n^n}{(n+1)^{n+1}}.$ Thus the maximum value is $\frac{n^n}{(n+1)^{n+1}}$ and it is attained when $\frac{x}{n} = 1-x \Leftrightarrow x = \frac{n}{n+1}$. Note that the same result will be obtained if we differentiated $x^n(1-x)$.

(c) (i)	$\sqrt{m+n} \leq \sqrt{m} + \sqrt{n}$ $\Leftrightarrow m+n \leq m+n+2\sqrt{mn}$ $\Leftrightarrow \sqrt{mn} \geq 0$ which is clearly true. Equality holds if and only if $m=0$ or $n=0$.
(ii)	We have $L = \int_a^b \sqrt{1+(f'(x))^2} \, dx \leq \int_a^b \sqrt{1} \, dx + \int_a^b \sqrt{(f'(x))^2} \, dx$ $= (b-a) + \int_a^b f'(x) \, dx = (b-a) + [f(x)]_a^b \quad \text{since } f'(x) > 0$ $= (b-a) + (f(b) - f(a)).$ Since $f_n(x) = x^n$ is increasing on $[0, 1]$ and $f(0) = 0$, $f(1) = 1$, we have $L_n \leq (1-0) + (1-0) = 2.$ Note that the inequality is actually strict as from (c)(i) we know that equality can hold only when either 1 or $f_n'(x) = nx^{n-1} = 0$ (identically) which is impossible.

2 (i)	Since $P_2'(x) = 2P_1(x) = 2x-1$, integrating gives $P_2(x) = x^2 - x + C$ for some constant C. Observe that the condition $P_2(0) = P_2(1)$ gives no information about C. Using $P_3'(x) = 3P_2(x) = 3x^2 - 3x + 3C$, gives $P_3(x) = x^3 - \frac{3}{2}x^2 + 3Cx + D$ for some constant D. Hence $P_3(0) = P_3(1)$ gives $D = 1 - \frac{3}{2} + 3C + D \Rightarrow C = \frac{1}{6},$ and so $P_2(x) = x^2 - x + \frac{1}{6}$.
(ii)	Continuing as in (i), we get $P_4'(x) = 4x^3 - 6x^2 + 2x + D$ $\Rightarrow P_4(x) = x^4 - 2x^3 + x^2 + Dx + E$ for some constant E. Hence $P_4(0) = P_4(1)$ gives $E = 1 - 2 + 1 + D + E \Rightarrow D = 0,$ and so $P_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$. Similarly, $P_5'(x) = 5x^4 - 10x^3 + 5x^2 + 5E$ $\Rightarrow P_5(x) = x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 + 5Ex + F$ for some constant F. Now $P_5(0) = P_5(1)$ gives $E = -\frac{1}{30}$ and $P_4(x) = x^4 - 2x^3 + x^2 - \frac{1}{30}$.

(iii)	Let $Q(n)$ denote the statement $P_{n+1}(x+1) = P_{n+1}(x) + (n+1)x^n$ for $n \geq 0$. When $n = 0$, it is easily seen that $\begin{aligned} P_{0+1}(x+1) &= (x+1) - \frac{1}{2} \\ &= \left(x - \frac{1}{2}\right) + 1 \\ &= P_{0+1}(x) + (0+1)x^0. \end{aligned}$ Thus the base case is true. Assuming $Q(k)$ is true for some k, we have $P_{k+1}(x+1) = P_{k+1}(x) + (k+1)x^k.$ Multiplying by $(k+2)$ gives $\begin{aligned} (k+2)P_{k+1}(x+1) &= (k+2)P_{k+1}(x) + (k+2)(k+1)x^k \\ \Rightarrow \frac{d}{dx}[P_{k+2}(x+1)] &= \frac{d}{dx}[P_{k+2}(x)] + (k+2)(k+1)x^k \\ \Rightarrow P_{k+2}(x+1) &= P_{k+2}(x) + (k+2)x^{k+1} + G, \end{aligned}$ where G is the constant of integration. Substituting $x = 0$ gives $P_{k+2}(1) = P_{k+2}(0) + G \Rightarrow G = 0$ since $P_{k+2}(0) = P_{k+2}(1)$. Hence $Q(k+1)$ is true. As $Q(0)$ is also true, $Q(n)$ is true for all $n \geq 0$ by induction.
(iv)	Rearranging the result in (iii) gives $x^n = \frac{1}{n+1} [P_{n+1}(x+1) - P_{n+1}(x)].$ Summing over x from 0 to m, with n constant results in $\begin{aligned} \sum_{x=0}^m x^n &= \frac{1}{n+1} \sum_{x=0}^m [P_{n+1}(x+1) - P_{n+1}(x)] \\ &= \frac{1}{n+1} \left[\sum_{x=0}^m P_{n+1}(x+1) - \sum_{x=0}^m P_{n+1}(x) \right] \\ &= \frac{1}{n+1} \left[\sum_{x=1}^{m+1} P_{n+1}(x) - \sum_{x=0}^m P_{n+1}(x) \right] \\ &= \frac{1}{n+1} [P_{n+1}(m+1) - P_{n+1}(0)]. \end{aligned}$ When $n = 3$, we get $\begin{aligned} \sum_{x=0}^m x^3 &= \frac{1}{4} [P_4(m+1) - P_4(0)] \\ &= \frac{1}{4} [(m+1)^4 - 2(m+1)^3 + (m+1)^2] \\ &= \frac{1}{4} (m+1)^2 [(m^2 + 2m + 1) - 2(m+1) + 1] \\ &= \frac{1}{4} m^2 (m+1)^2 \end{aligned}$ using the formula for $P_4(x)$ from (ii).

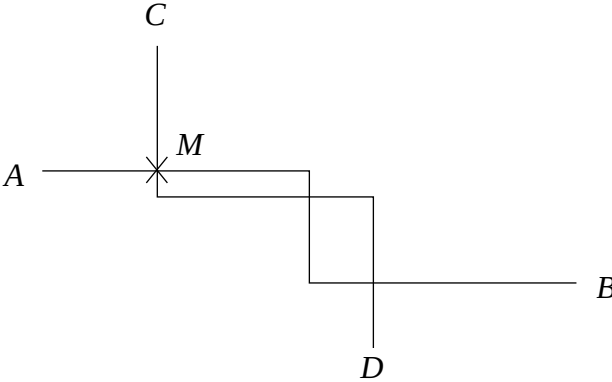
3(i)	Using a substitution of $x = a - y$, and the given $f(x) = f(a - x)$, $\int_0^a xf(x) dx = \int_a^0 (a - y)f(a - y) (-1) dy$ $= \int_0^a af(a - y) dy - \int_0^a yf(a - y) dy$ $= \int_0^a af(a - y) dy - \int_0^a xf(x) dx.$ Bringing the second integral of the RHS to the LHS will give $\int_0^a xf(x) dx = \frac{a}{2} \int_0^a f(a - x) dx.$
(ii)	Immediate application of part (i) using $a = \pi$ and $f(x) = \sin^n x$, since $\sin^n x = \sin^n(\pi - x)$, gives the first equality $\int_0^\pi x \sin^n x dx = \frac{\pi}{2} \int_0^\pi \sin^n x dx$. For the second equality, by symmetry of the function about $x = \frac{\pi}{2}$, or otherwise again an application of a substitution, i.e, $\frac{\pi}{2} \int_0^\pi \sin^n x dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin^n x dx + \frac{\pi}{2} \int_{\frac{\pi}{2}}^\pi \sin^n x dx$ $= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin^n x dx + \frac{\pi}{2} \int_{\frac{\pi}{2}}^0 -\sin^n(\pi - y) dy$ $= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin^n x dx + \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin^n x dx = \pi \int_0^{\frac{\pi}{2}} \sin^n x dx.$
(iii)	Using integration by parts, $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \sin x \sin^{n-1} x dx$ $= -\cos x \sin^{n-1} x \Big _0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \cos^2 x \sin^{n-2} x dx$ $= \int_0^{\frac{\pi}{2}} (n-1)(1 - \sin^2 x) \sin^{n-2} x dx$ $= \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x dx - \int_0^{\frac{\pi}{2}} (n-1) \sin^n x dx$ Bring the RHS last term to the LHS, we obtain $I_n + (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} (n-1) \sin^{n-2} x dx \Rightarrow nI_n = (n-1)I_{n-2}.$
(iv)	Using part (ii) and (iii), with $n = 11$, $\int_0^\pi x \sin^{11} x dx = \pi \int_0^{\frac{\pi}{2}} \sin^{11} x dx$ $= \pi I_{11} = \pi \frac{10}{11} I_9 = \dots = \pi \frac{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10}{3 \cdot 5 \cdot 7 \cdot 9 \cdot 11} I_1 = \frac{256\pi}{693}.$ Since $I_1 = \int_0^{\frac{\pi}{2}} \sin x dx = -\cos x \Big _0^{\frac{\pi}{2}} = 1$.
(v)	Multiplying I_{n-1} to the recurrence in part (iii) gives $x_n = nI_n I_{n-1} = (n-1)I_{n-1} I_{n-2} = x_{n-1}$. Hence it is a constant sequence that equals to $x_1 = I_1 I_0 = \frac{\pi}{2}$ for every term of the sequence.

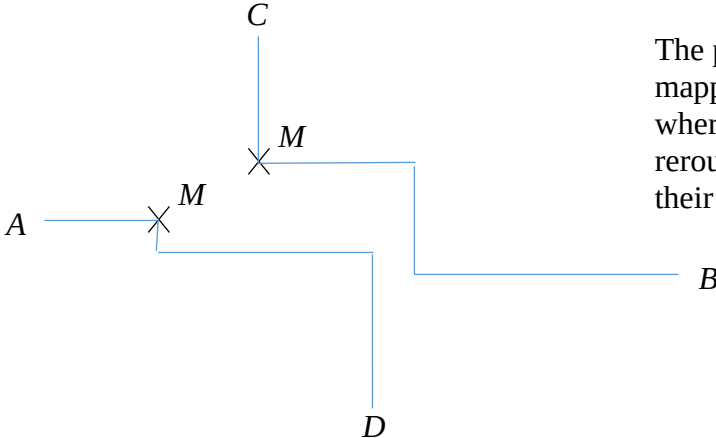
(vi)	From part (v), it is clear that $I_n I_{n-1} = \frac{x_n}{n} = \frac{\pi}{2n}$. $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx = \int_0^{\frac{\pi}{2}} \sin x \sin^{n-1} x \, dx \leq \int_0^{\frac{\pi}{2}} \sin^{n-1} x \, dx = I_{n-1}$ since $0 \leq \sin x \leq 1$, so I_n is a decreasing sequence of positive numbers. Hence $\frac{\pi}{2n} = I_n I_{n-1} \geq I_n I_n \Rightarrow I_n \leq \sqrt{\frac{\pi}{2n}}$. Similarly, $\frac{\pi}{2(n+1)} = I_{n+1} I_n \leq I_n I_n \Rightarrow I_n \geq \sqrt{\frac{\pi}{2(n+1)}}$.
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4(a)(i)	We are finding the number of integer solutions to the equation $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 42$ where x_i for $i = 1$ to 6 represents the number of sweets each friend has (in order A to F). The condition in the question says that $x_1 \geq 3, x_2 \geq 4, x_3 \geq 5$. This problem is the same as distributing 42 identical balls into 6 distinct boxes satisfying the above conditions. To meet the requirements, 3, 4, and 5 ($3 + 4 + 5 = 12$) balls can be placed into the first three boxes respectively. The total is now 30. To calculate the above we need the total number of ways for 30 identical balls to be placed into six distinct boxes which is $\binom{30+6-1}{6-1} = \binom{35}{5} = 324632$.
(a)(ii)	For Denny, Elly and Fanny's request on top of the earlier 3, we let A be the ways to put 30 identical balls into the six distinct boxes such that $x_4 > 8$, B be the ways to put 30 identical balls into the six distinct boxes such that $x_5 > 8$ and C be the ways to put 30 identical balls into the six distinct boxes such that $x_6 > 8$. Clearly, the desired answer is $A' \cap B' \cap C'$. To calculate A, first place 9 balls into fourth box, and then distribute 21 identical balls into six distinct boxes, which gives $\binom{21+6-1}{6-1} = \binom{26}{5} = 65780$. Similarly for B and C. Next to calculate $A \cap B$, place 9 balls into the fourth and fifth boxes each, and the remaining 12 identical balls into the six distinct boxes, which gives $\binom{12+6-1}{6-1} = \binom{17}{5} = 6188$. Similarly for $B \cap C$ and $A \cap C$. Lastly for $A \cap B \cap C$, place 9 balls each into the fourth, fifth and sixth boxes, and there are 3 identical balls left to put into the six distinct boxes and the number of ways is $\binom{8}{5} = 56$. By PIE, the total number of ways is $324632 - 3(65780) + 3(6188) - 56 = 145800.$
(b)(i)	Assume boxes are distinct, so we are calculating $2S(r, 2)$. There are two ways to distribute each object into either box, so 2^r. Since no box is empty we remove the two cases where either box is empty. Hence $2S(r, 2) = 2^r - 2$ $S(r, 2) = 2^{r-1} - 1.$

(ii)	Let a be a particular object among the r distinct objects. In any way of distributing the r distinct objects into n identical boxes such that no box is empty, either (i) a is alone in one box or (ii) a is not alone in one box In case (i), there remains $(r - 1)$ distinct objects to be distributed into $(n - 1)$ identical boxes with no empty boxes. The number of ways to do this is $S(r - 1, n - 1)$. In case (ii), the $(r - 1)$ distinct objects (excluding a) are first put into the n identical boxes such that no box is empty. This can be done in $S(r - 1, n)$ ways. Object a can then be placed in any of the boxes in n ways. By (MP), number of ways in case (ii) is $n S(r - 1, n)$. Thus, by (AP), $S(r, n) = S(r - 1, n - 1) + n S(r - 1, n)$ for $r \geq n + 1$.
(iii)	By use of parts (i) and (ii), we have $S(5, 3) + S(5, 2) + S(5, 1)$ $= S(4, 2) + 3S(4, 3) + S(5, 2) + S(5, 1)$ $= S(4, 2) + 3[S(3, 2) + 3S(3, 3)] + S(5, 2) + S(5, 1)$ $= 2^{4-1} - 1 + 3[2^{3-1} - 1 + 3] + 2^{5-1} - 1 + 1$ $= 41.$

5 (i)	The number of compositions is the same as the number of ways of distributing n identical objects into 1, 2, 3, ... or n distinct boxes with no box empty. To put n identical objects into k distinct boxes where $1 \leq k \leq n$, one object is placed into each box, and the remaining $n - k$ balls are placed into the k distinct boxes, which is a bijection between the set of ways of distributing the balls and the set of $n - k + k - 1 = n - 1$ binary sequences with $k - 1$ ones. Hence by BP, we have $\binom{n-1}{k-1}$. By AP, the number of boxes can be from 1 to n, so the total number of compositions is $\sum_{k=1}^n \binom{n-1}{k-1}$.
(b)(i)	Using the same idea as part (i), except that we need to place 2 identical balls into each distinct box (to ensure that each box has 2 or more balls, i.e., each summand is at least 2), hence the summation takes from 1 to $\left\lfloor \frac{n+1}{2} \right\rfloor$ steps (or 1 to $\left\lfloor \frac{n+1}{2} \right\rfloor$ distinct boxes), and the bijection is from the set of ways of distributing the remaining $n + 1 - 2k$ identical balls into k distinct boxes to the set of $(n + 1 - 2k) + (k - 1)$-digit binary sequences with $k - 1$ ones, is $\binom{(n+1-2k)+(k-1)}{k-1} = \binom{n-k}{k-1}$.
(ii)	Consider F_{n+1}, $n \geq 2$, which is the number of compositions for $n + 2$ using integers larger than one. Method 1 Consider the first term in the composition. It is either 2 or more than 2. If it is 2, then the remaining integers form a composition of n. If it is more than 2, subtract 1 from it, and it is a composition of $n + 1$. Hence we have $F_{n+1} = F_n + F_{n-1}$. Method 2 There is one composition with only one term that is $n + 2$ itself. For other compositions, let the first term be k, where $2 \leq k \leq n$. By AP, $F_{n+1} = 1 + F_1 + \dots + F_{n-1}$. Then this also yields $F_n = 1 + F_1 + \dots + F_{n-2}$, and on combining this two equations, we have $F_{n+1} = F_n + F_{n-1}$.

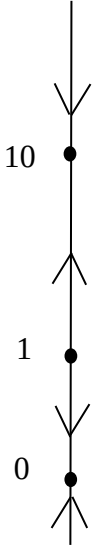
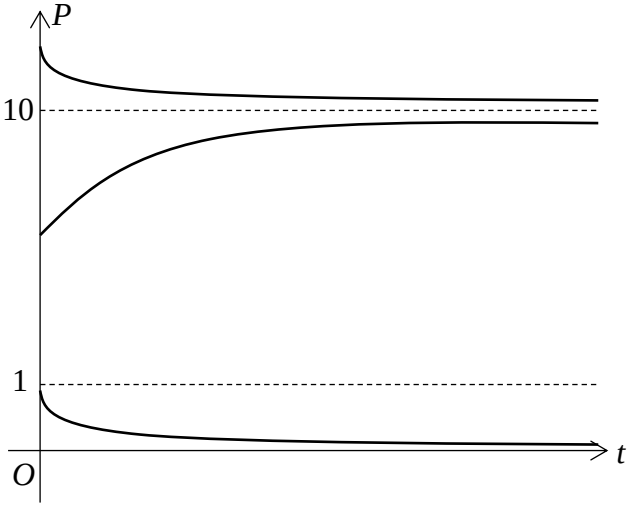
6 (a) (i)	Let X be the set of shortest routes from A to B. Let Y be the set of sequences of 7E and 3S. Define a bijection f from X to Y in the following manner: We write E if we move east and S if we move south. Clearly, every sequence of 7E and 3S correspond to a shortest route from A to B. Hence f is surjective. Likewise, for two identical permutations of 7E and 3S, they must correspond to the same shortest route. Hence f is injective and thus bijective. Thus $s(AB) = \binom{10}{3}$.
(ii)	Number of shortest routes $= \binom{10}{3} - \binom{5}{1}\binom{5}{2} = 70$.
(b) (i)	We need to check for solutions to $120 = \binom{x}{k}$, $2 \leq k \leq 8$. Using a GC, we have $\binom{16}{2} = \binom{10}{3} = \binom{10}{7} = 120$. There are 3 therefore cases. <u>Case 1:</u> 7E and 3S Only 3 possibilities here, with A at $(-4, 2)$, $(-3, 1)$ or $(-3, 2)$ <u>Case 2:</u> 7S and 3E There are 12 possibilities here, with A at (x, y), where $-2 \leq x \leq -1$, $1 \leq y \leq 6$. <u>Case 3:</u> 14S and 2E Only 3 possibilities here, with A at $(-1, 6)$, $(-1, 7)$ or $(-1, 8)$. Hence a total of 18 such ways.
(b) (ii)	Consider the set X of all possible pairs of routes from A to B and from C to D, and the set Y of all possible pairs of routes from A to D and from C to B. We will show that there is an injective map from X to Y. We will define the injection f as follows. Consider an arbitrary pair of routes from X. Clearly, these routes will intersect, say at point M. Reroute the path from A to B to a path to D by changing its course at M. Likewise, reroute the path from C to D to a path to B by changing its course at M. Hence we obtain a pair of routes from Y. This map is injective by definition. Hence $s(AB)s(CD) \leq s(AD)s(BC)$. To see why the inequality is strict, it suffices to show that there is a pair of routes from Y that does not have a pre-image. Clearly, the path that goes directly south from A before directly east to D and the path that goes directly east from C before directly south to B do not intersect and thus does not have a pre-image. Hence $s(AB)s(CD) < s(AD)s(BC)$.  A pair of routes from A to B and C to D intersecting at M.

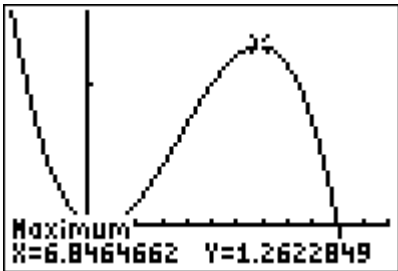
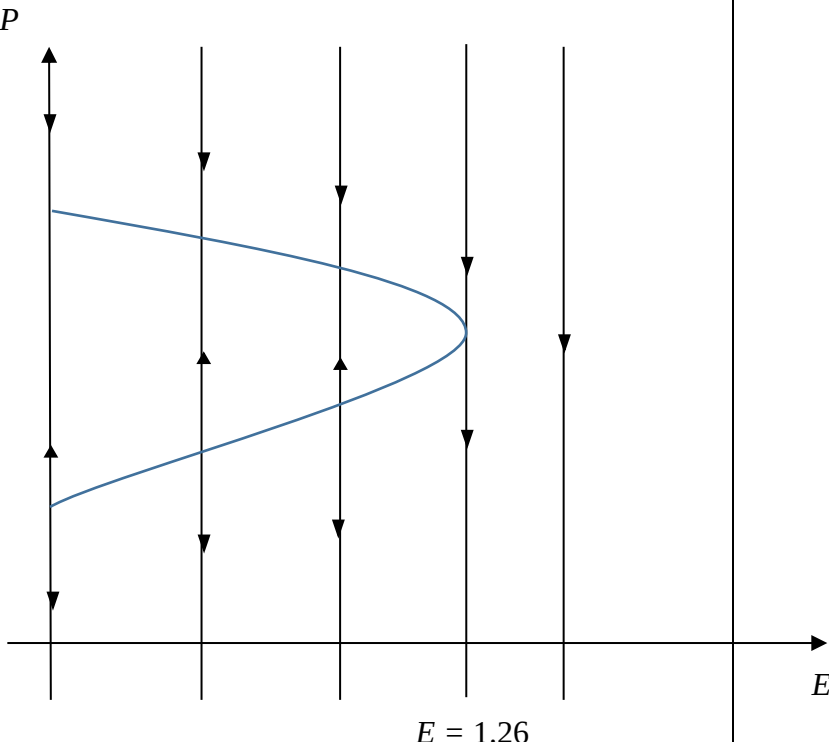
	 The pair of routes above is then mapped to the following pair, where the original paths are rerouted to take the other path at their first point of intersection, M.
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7(i)	$2 \frac{d^2 x}{dt^2} + 8 \frac{dx}{dt} + cx = 0$ $2(8e^{-2t}) + 8(-4e^{-2t}) + c(2e^{-2t}) = 0$ $2c = 16$ $c = 8$
(ii)	$2 \frac{d^2 x}{dt^2} + 8 \frac{dx}{dt} + 8x = 0$ $\frac{d^2 x}{dt^2} + 4 \frac{dx}{dt} + 4x = 0$ $r^2 + 4r + 4 = 0$ $r = -2$ Both particles experience critical damping.
(iii)	$x = (At + B)e^{-2t}$ $3 = x(0) = B$ $\frac{dx}{dt} = e^{-2t} (A - 2(At + 3))$ $\frac{dx}{dt} \Big _{t=0} = A - 6 = 0 \Rightarrow A = 6$ $x = (6t + 3)e^{-2t}$ Hence when $t = 3$, $x = 21e^{-6}$.

8(a)	Integrating factor $I(x) = e^{\int \frac{-2x}{x^2+1} dx} = e^{-\ln(x^2+1)} = \frac{1}{x^2+1}$. $\therefore \frac{d}{dx} \left(\frac{y}{x^2+1} \right) = \frac{1}{x^2+1}$ $\frac{y}{x^2+1} = \int \frac{1}{x^2+1} dx = \tan^{-1} x + C$ $y = (x^2+1)(\tan^{-1} x + C)$
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(b) (i)	With $x = \cos \theta$, we have $\frac{dx}{d\theta} = -\sin \theta$ and $\frac{d\theta}{dx} = -\frac{1}{\sin \theta}$. Therefore $\begin{aligned}\frac{dy}{dx} &= \frac{dy}{d\theta} \frac{d\theta}{dx} && \text{(Chain rule)} \\ &= -\frac{1}{\sin \theta} \frac{dy}{d\theta}\end{aligned}$ and $\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) \\ &= \frac{d}{d\theta} \left(\frac{dy}{dx} \right) \frac{d\theta}{dx} && \text{(Chain rule again)} \\ &= -\frac{1}{\sin \theta} \frac{d}{d\theta} \left(-\frac{1}{\sin \theta} \frac{dy}{d\theta} \right) \\ &= \frac{1}{\sin \theta} \left(\frac{1}{\sin \theta} \frac{d^2y}{d\theta^2} - \frac{\cos \theta}{\sin^2 \theta} \frac{dy}{d\theta} \right) \\ &= \frac{1}{\sin^2 \theta} \left(\frac{d^2y}{d\theta^2} - \frac{\cos \theta}{\sin \theta} \frac{dy}{d\theta} \right).\end{aligned}$ Using these in (1), and noting $1 - x^2 = 1 - \cos^2 \theta = \sin^2 \theta$, we get $\begin{aligned}(1 - x^2) \left[\frac{1}{\sin^2 \theta} \left(\frac{d^2y}{d\theta^2} - \frac{\cos \theta}{\sin \theta} \frac{dy}{d\theta} \right) \right] - x \left[-\frac{1}{\sin \theta} \frac{dy}{d\theta} \right] + p^2 y &= 0 \\ \Rightarrow \frac{d^2y}{d\theta^2} + p^2 y &= 0.\end{aligned}$
(ii)	Characteristic equation: $r^2 + p^2 = 0 \Rightarrow r = \pm pi$. Therefore general solution is $\begin{aligned}y &= A \cos p\theta + B \sin p\theta \\ &= A \cos(p \cos^{-1} x) + B \sin(p \cos^{-1} x),\end{aligned}$ where A and B are arbitrary constants.
(iii)	With $p = 3$, $y = A \cos(3 \cos^{-1} x) + B \sin(3 \cos^{-1} x)$. When $x = 0$, $\cos^{-1} x = \cos^{-1} 0 = \frac{\pi}{2}$, so this gives $y = -B$. Hence $B = 0$ and $y = A \cos(3 \cos^{-1} x).$ Using $\cos 3\alpha = 4 \cos^3 \alpha - 3 \cos \alpha$, we get $y = A(4x^3 - 3x)$. Therefore $\frac{dy}{dx} = A(12x^2 - 3)$. At $x = 0$ this gives $\frac{dy}{dx} = -3A$, so $A = 1$.

9 (i)	If $P < 1$ then $\frac{dP}{dt} = \frac{1}{10} P \left(1 - \frac{P}{10} \right) (P - 1) < 0$
(ii)	Carrying capacity is 10.
(iii)	0, 1, 10 are the equilibrium points 0 and 10 are stable while 1 is unstable 
(iv)	
(b)(i)	$\frac{dP}{dt} = \frac{1}{10} P \left(1 - \frac{P}{10} \right) (P - 1) - E$

(ii)	 Maximum X=6.8464662 Y=1.2622849 The bifurcation value is 1.26.  $E = 1.26$																				
(iii)	<table><tr><th>t</th><th>P</th><th>$\frac{1}{10}P\left(1-\frac{P}{10}\right)(P-1)$</th><th>$\tilde{P}$</th><th>$\frac{\Delta P}{\Delta t}$</th></tr><tr><td>0</td><td>2</td><td>0.16</td><td>2.08</td><td>0.16896</td></tr><tr><td>0.5</td><td>2.08448</td><td>0.17894</td><td>2.17395</td><td>0.18933</td></tr><tr><td>1</td><td>2.17915</td><td></td><td></td><td></td></tr></table>	t	P	$\frac{1}{10}P\left(1-\frac{P}{10}\right)(P-1)$	$\tilde{P}$	$\frac{\Delta P}{\Delta t}$	0	2	0.16	2.08	0.16896	0.5	2.08448	0.17894	2.17395	0.18933	1	2.17915			
t	P	$\frac{1}{10}P\left(1-\frac{P}{10}\right)(P-1)$	$\tilde{P}$	$\frac{\Delta P}{\Delta t}$																	
0	2	0.16	2.08	0.16896																	
0.5	2.08448	0.17894	2.17395	0.18933																	
1	2.17915																				
(iv)	Use a smaller time step.																				